Exercise 1: Stress Analysis

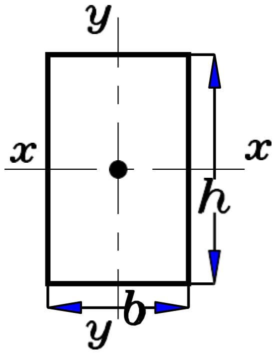
Geometrical Properties of Plane Sections

Rectangle

$$A = b \times h$$

$$I_x = \frac{b \times h^3}{12}$$

$$I_y = \frac{h \times b^3}{12}$$

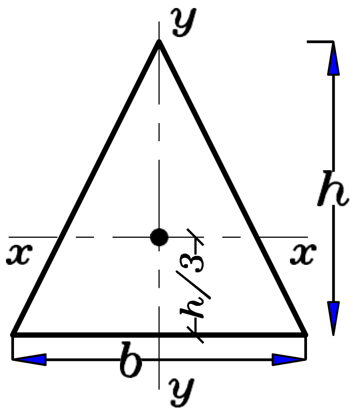


Triangle

$$A = \frac{1}{2} \times b \times h$$

$$I_x = \frac{b \times h^3}{36}$$

$$I_y = \frac{h \times b^3}{48}$$

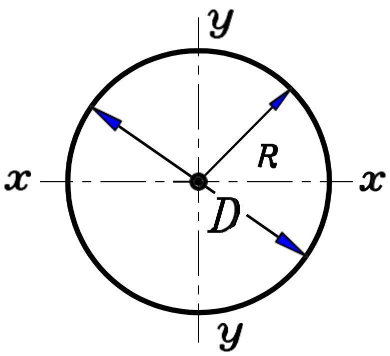


Circle

$$A = \frac{\pi D^2}{4}$$

$$I_x = I_y = I = \frac{\pi D^4}{64}$$

$$J = \frac{\pi D^4}{32}$$

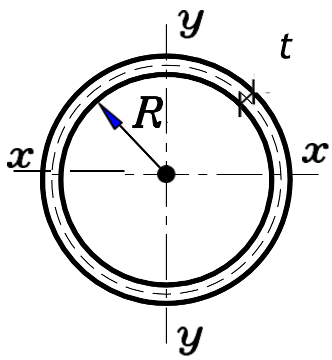


Thin Tube

$$A = 2\pi Rt = \pi Dt$$

$$J = 2I = 2\pi R^3 t$$

$$I_x = I_y = I = \pi R_3 t$$

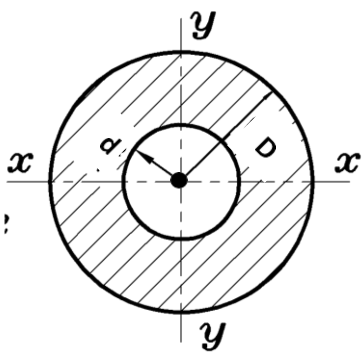


Hollow Circle

$$A = \frac{\pi}{4}(D^2 - d^2)$$

$$I = \frac{\pi}{64}(D^4 - d^4)$$

$$J = \frac{\pi}{32}(D^4 - d^4)$$

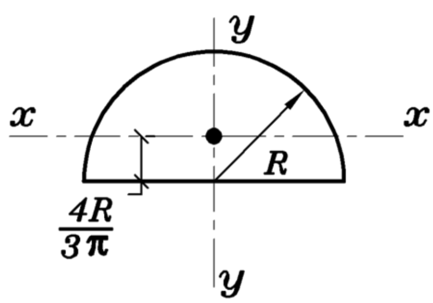


Semi-Circle

$$A = \frac{\pi d^2}{8}$$

$$I_x = 0.11 \times R^4$$

$$I_y = \frac{\pi}{8} \times R^4 = 0.3926 R^4$$



Note:

A: Area

I : Moment of Inertia

J : Polar Moment of Inertia

In General:

$$J = I_x + I_y$$

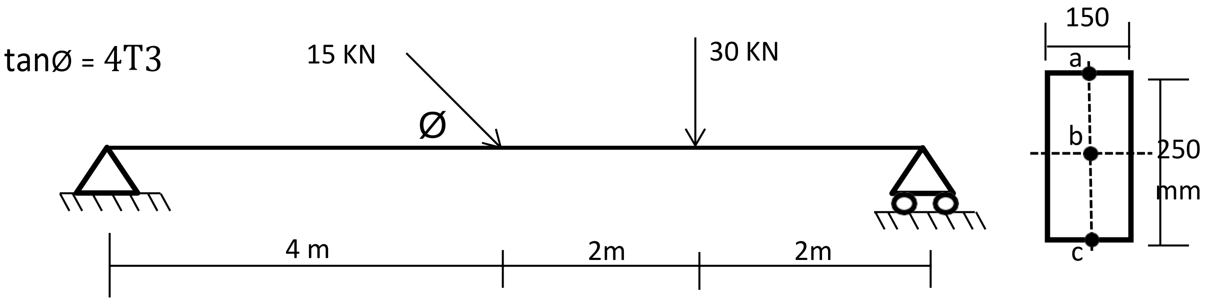
For circular cross section $I_x = I_y = I \quad \therefore J = 2I$

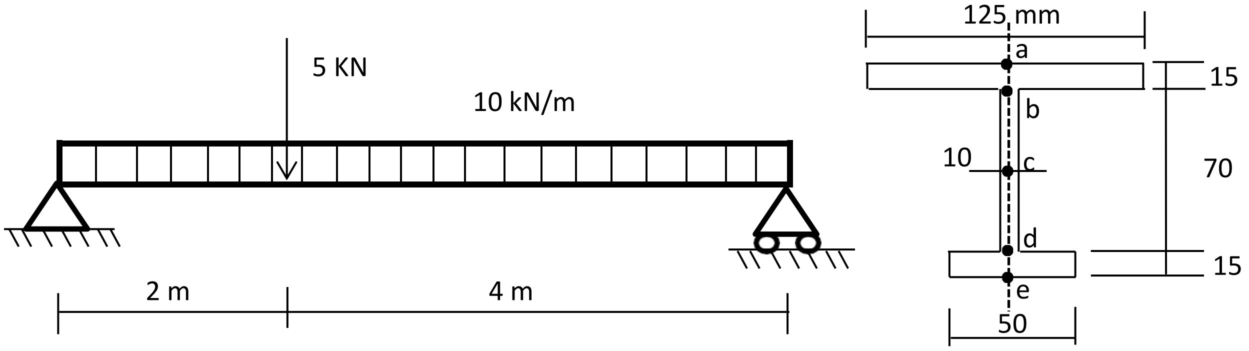
1.

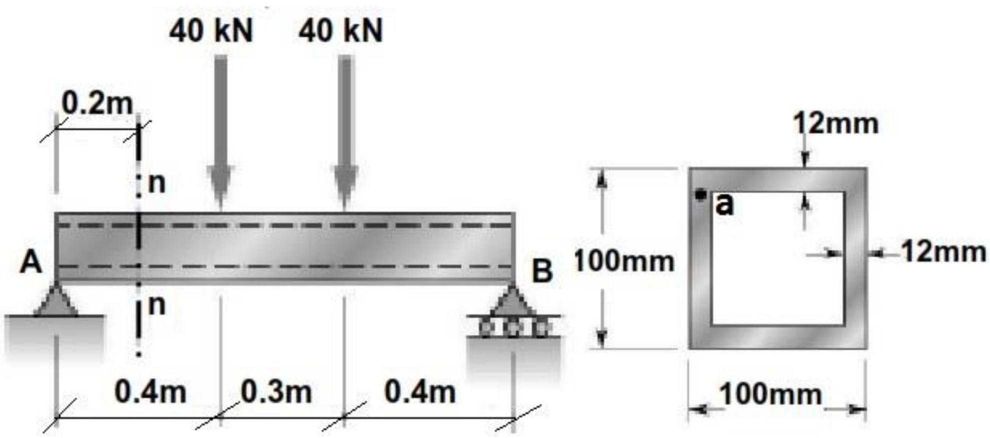
a. **Draw** the normal force, the shearing force, and the bending moment diagrams for each of the following structures.

b. **Calculate** and draw the normal and shear stress diagrams at the critical sections for the given cross sections.

c. **Draw** the state of stress on an infinitesimal element for the shown points **a**, **b**, **c**, **d** and **e**.





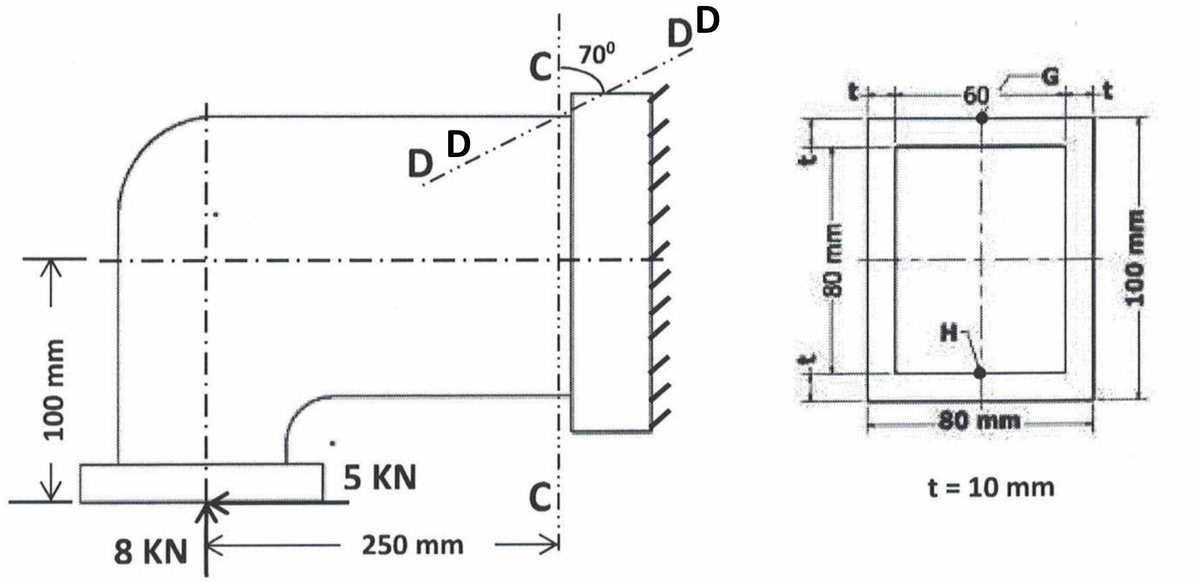


2. The shown frame is subjected to the given loads and has the shown cross section.

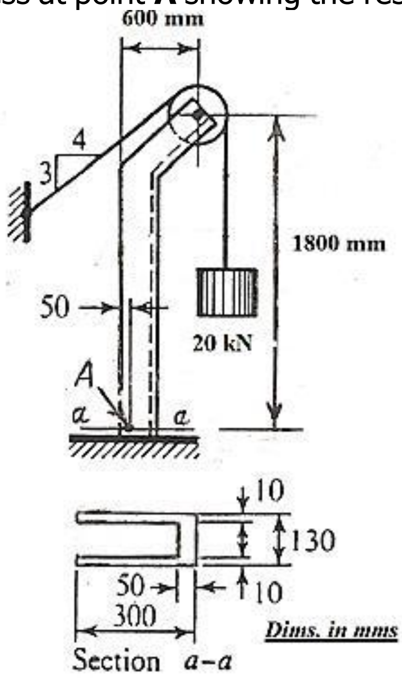
a. **Calculate** and draw the normal and shear stress diagrams at section **C-C**.

b. **Determine** the state of stresses at points **G** and **H** showing the results on infinitesimal elements.

c. **Calculate** the normal and shear stresses on the inclined plane **D-D** at point **G**.

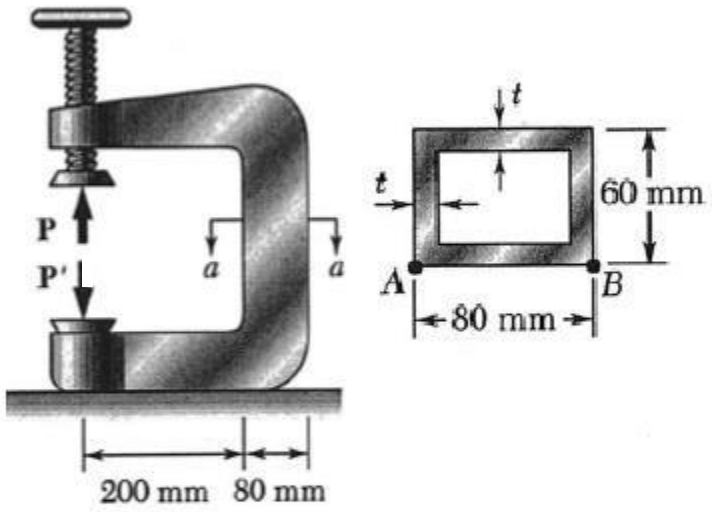


3. A special hoist is loaded with a **20 kN** load suspended by a cable as shown.
Calculate and draw the normal and shear stress diagrams at section **a-a**.
Determine the state of stress at point **A** showing the results on an infinitesimal element.



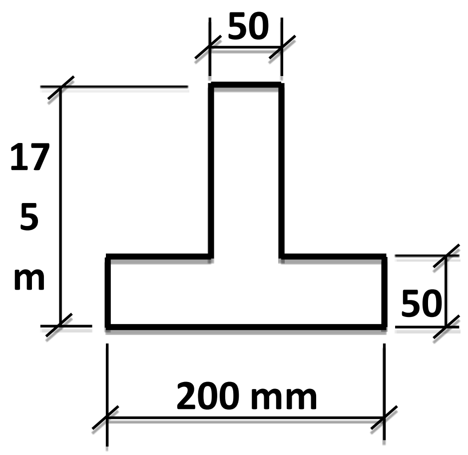
4. The vertical portion of the press shown in Figure is a rectangular tube having a wall thickness of **t= 10mm**. If the force exerted by the press (**P**) is **30 kN**;

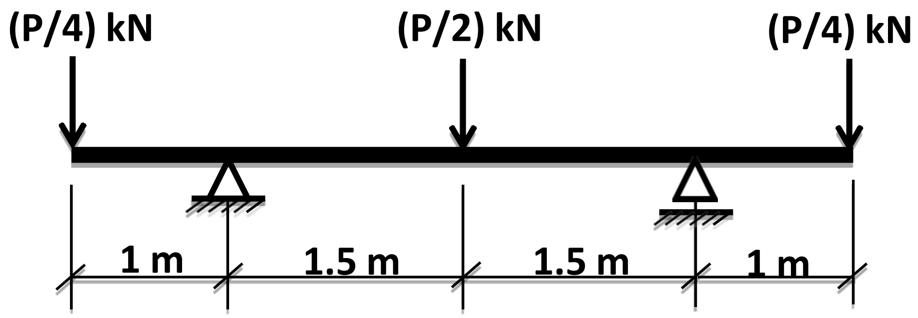
Draw the normal stress distributions in section a-a and determine the maximum tensile and compressive stresses acting in the section **a-a**, Showing the result on infinitesimal element.



5. The shown beam with overhanging ends is loaded by three concentrated forces. The beam is simply supported and has the shown cross-section .The beam material is grey cast iron having an allowable working stress in tension of **35 MPa** and in compression of **150MPa**.

Determine the allowable value of **P**.(**Neglect** the effect of Transverse Shear)

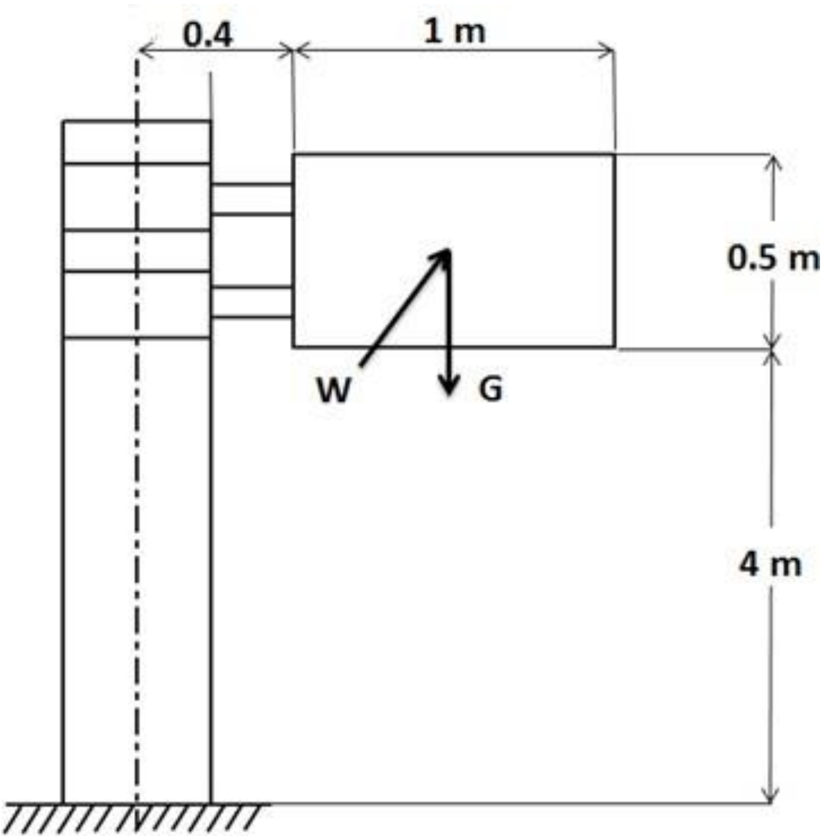




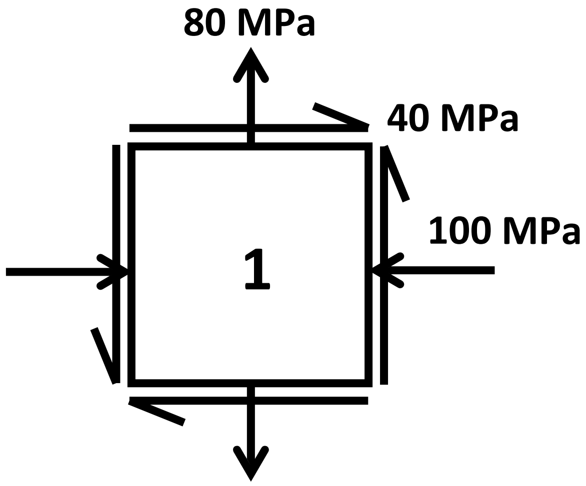
6. The shown sign post is fixed to a steel pipe **100 mm** outside diameter and **6** mm thick. Consider the own weight of the sign (**G**) as 1000 N and the maximum horizontal wind pressure on the sign (**W**) as **1200 N/m²**.

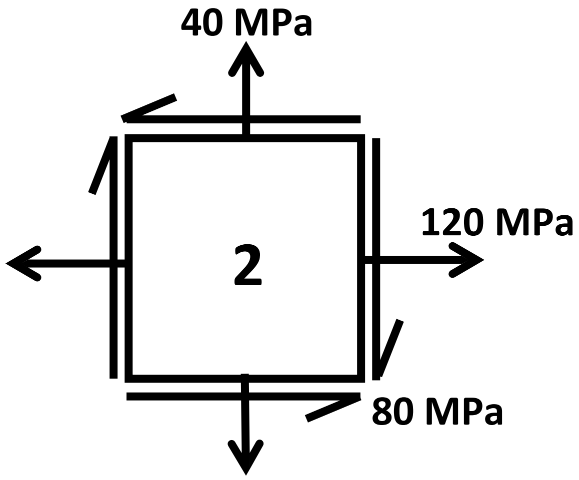
Calculate and draw the normal and shear stress diagrams **at the section of fixation**. At the points of maximum tensile and compressive stresses **draw** the state of stresses on infinitesimal elements, then find the values of maximum principal shear stresses at these points.

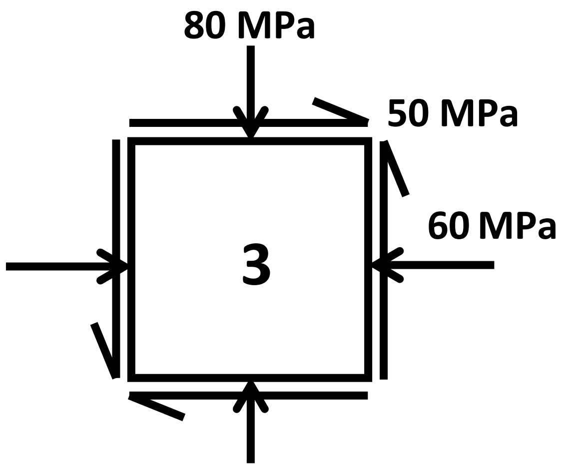
(**Neglect** the effect of transverse shear and the effect of normal stress due to normal force)

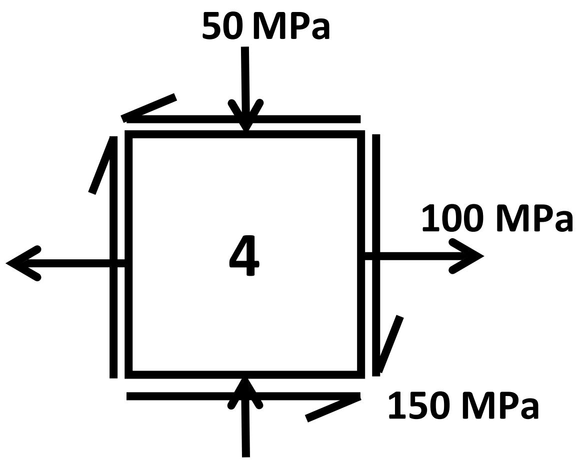


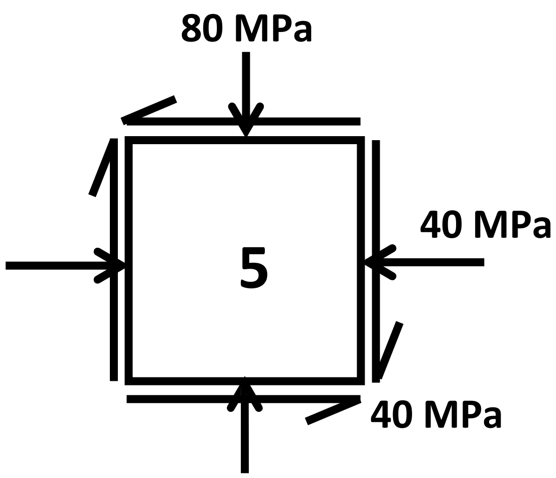
7. For the given elements, find analytically and by Mohr's Circle the principle normal and shear stresses and determine their planes.



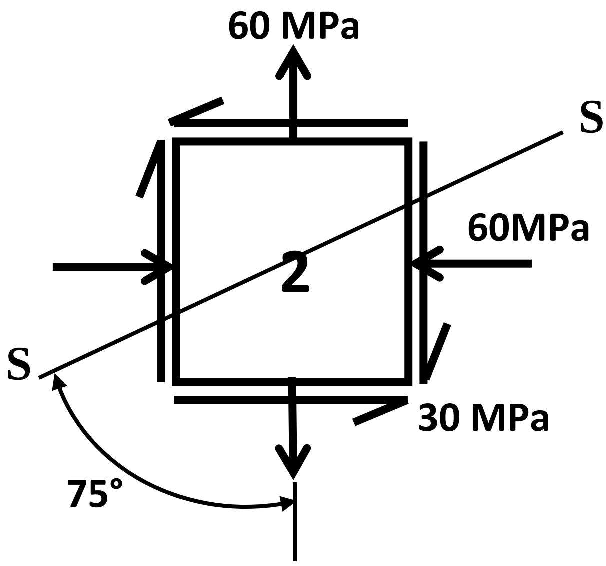


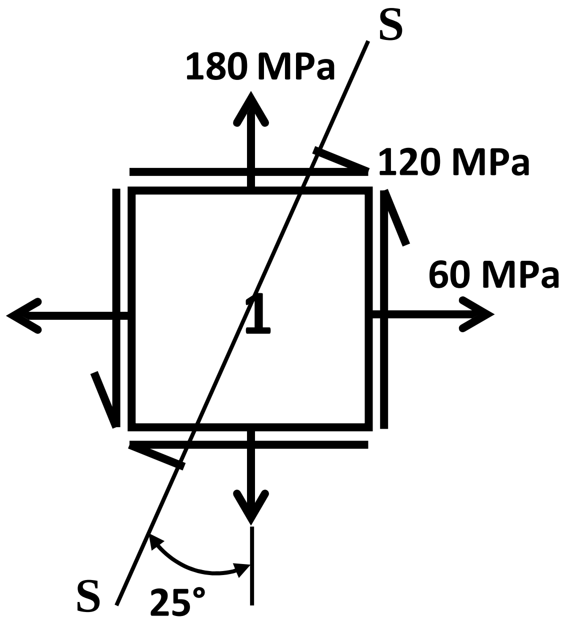


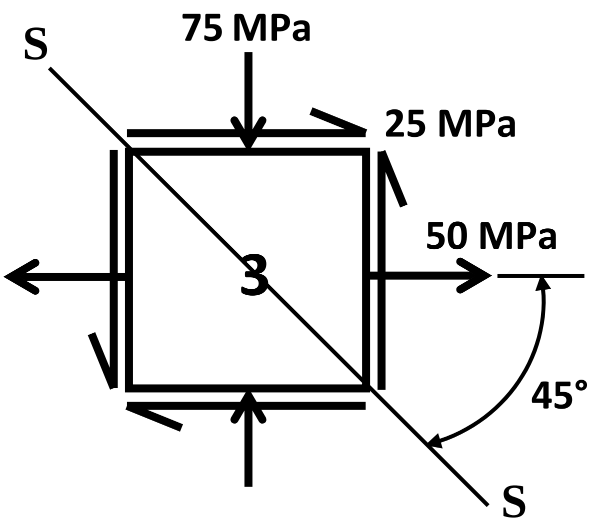




8. For the given state of stress elements, determine the state of stress at plane s-s.





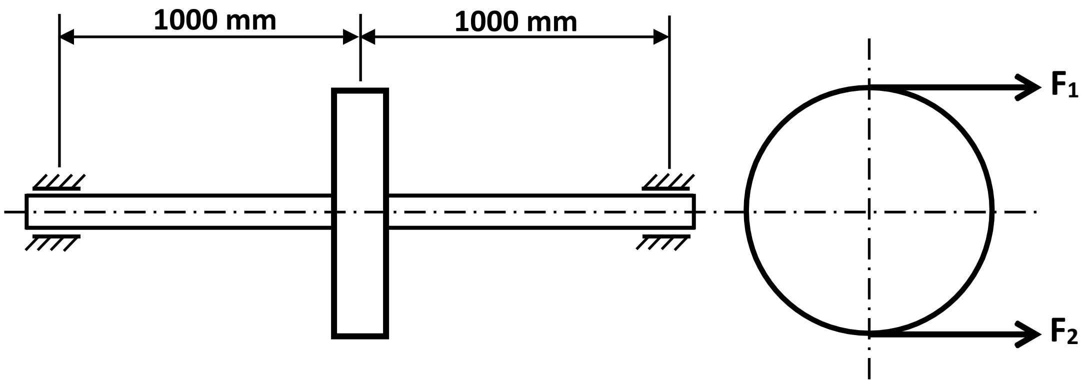


9. A sectional commercial shaft **2 m** long between bearings carries **1 kN** pulley at its midpoint, as shown in the figure. The pulley is keyed to the shaft and receives **30 kW** at **150 rpm** which is transmitted to a flexible coupling just outside the right bearing. The belt drive is horizontal and the sum of the belt tension is **8 kN**.

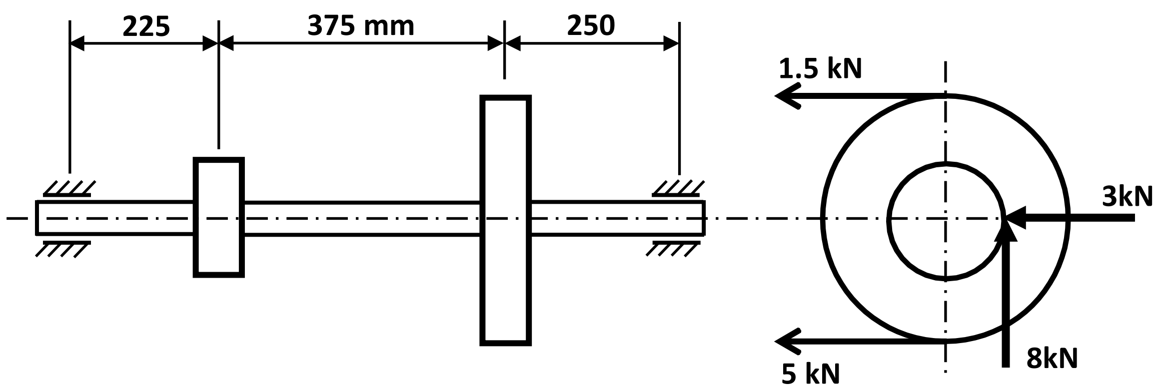
a. Calculate the shaft diameter.

b. Determine the angle of twist between bearings.

(Given that k_b=k_t=1.5 , G = 80 GPa , τ_{all} = 40 MPa)



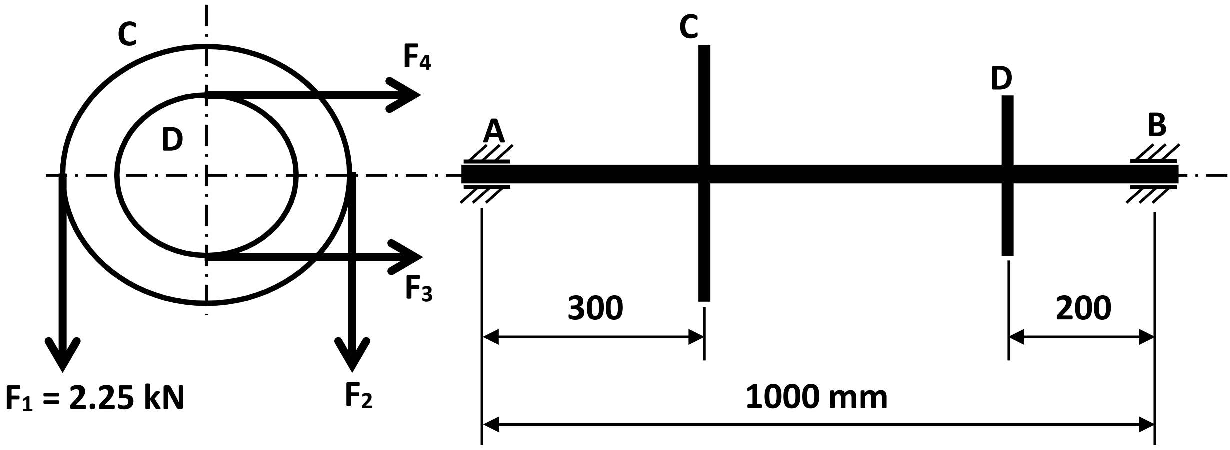
10. A **600 mm** diameter pulley driven by a horizontal belt transmits power through a solid shaft to a pinion which drives a mating gear. The pulley weighs **1.2 kN** to provide some flywheel effect. The arrangement of elements, the belt tensions, and the components of the gear reactions on the pinion are as indicated in the figure below. **Determine** the necessary shaft diameter according to the maximum principle shear stress theory with safety factor = 2.75, shock factors of k_b = 2, k_t = 1.5 and shaft material is St42 (σ_y = 220 MPa)



11. A shaft is supported by two bearings (A & B) placed one meter apart. Pulley C (**600 mm** diameter) is mounted at a distance of **300 mm** to the right of left bearing and this drives a pulley directly below it with the help of the belt having maximum tension of **2.25 kN**. Pulley D (**400 mm** diameter) is placed **200 mm** to the left of the right bearing and is driven by the help of an electric motor and a belt, which is placed horizontally to the right. The angle of contact for both pulleys is **180°** and coefficient of friction **μ = 0.24**.

a. Determine the suitable shaft diameter according to the maximum shear stress theory with a safety factor of 2.6, combined fatigue and shock factors for bending and torsion may be taken as 1.5 and 2 respectively.
Shaft material is St42 (σ_u = 420 MPa and σ_y = 220 MPa, G = 80 GPa)

b. Calculate the angle of twist between bearings.



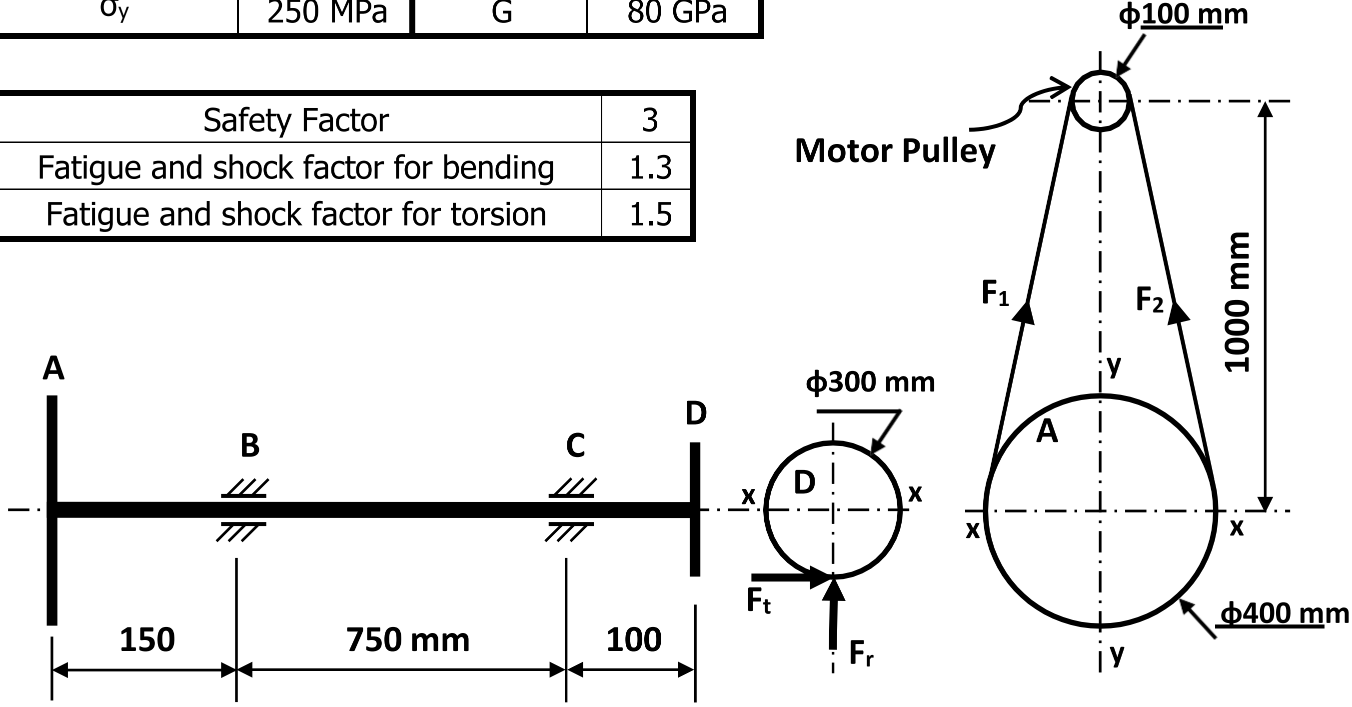
12. The transmission shaft ABCD transmits **18 kW** from pulley A (**φ400 mm**) to gear D (**φ300 mm**) at **160 rpm** and is supported on two bearings at B and C as shown. ($\mathbf{F_1 = 2.6\, F_2}$ & $\mathbf{F_r = 0.35\, F_t}$).

a. Determine the suitable shaft diameter according to the Maximum Principle Shear Stress Theory.

b. Calculate the angle of twist between bearings.

Shaft material St42			
σ_u	420MPa	E	210 GPa
σ_y	250 MPa	G	80 GPa

Safety Factor	3
Fatigue and shock factor for bending	1.3
Fatigue and shock factor for torsion	1.5



13. A shaft S is rotating at **500 rpm** is designed to transmit **7.5 kW** received from shaft **W**, which rotates at **1500 rpm**, through a **100 mm** diameter spur gear **K**. Power is delivered from shaft **S** to shaft **L** through a **100 mm** diameter spur gear **G**.

a. Determine the suitable shaft diameter for shaft S according to the Maximum Principle Shear Stress Theory.
Safety Factor = 3, combined fatigue and shock factors for bending $ torsion may be taken as 1.8 and 1.5 respectively. Shaft material is St42 (σ_u = **420 MPa** and σ_y = **250 b**).

b. Calculate the angle of twist between bearings A and B (E_{Steel} = **210 GPa** and v = **0.3**)